Asymptotic Matching and Near-NHEK QNMs

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This note describes the asymptotic matching procedure needed to construct the radial solution to the near extremal Teukolsky equation everywhere. A few things to keep in mind: These solutions should be valid everywhere for $r \in \mathbb{C}$. Additionally, the asymptotic matching requires that $\omega \to m/2$ as we near extremality. As such, we don't consider the negative frequency $m < 0$ modes. Theoretically, this could include the positive frequency $-m$ modes (mirror modes), but I haven't flushed that out yet. Then for $m = 0$, everything just breaks and things become singular. I think a slightly different method is needed for those modes. In it's application to higher derivative theories of [1], I think the only thing that changes is the angular separation constant $\lambda_{\ell m} \to \lambda_{\ell m} + \delta\lambda_{\ell m}$. However, the thing I'm not sure about is that in these theories, the horizon frequency changes, meaning that near extremality $\omega \nrightarrow m/2$. In [2], this doesn't seem to change the near NHEK Teukolsky equation, but I'm not sure how that's possible. This needs more investigation.

The logic of the work is as follows. Given the normal radial Teukolsky equation, we can take two limits, both while taking $a \to M$. The first limit is the inner region, where we zoom in near the inner and outer horizons. We can then solve this limit of the TE in the inner region. We can do the same thing in the outer region, getting an outer solution. The inner solution should be ingoing at the horizon and the outer solution should be outgoing at infinity. This information gives us a ratio for the coefficients of the linear combination of solutions that form the full outer solution. Then we also know that there is an intermediate zone where both the inner and outer solutions are valid. In this intermediate zone, the two solutions must match, giving us a second condition on the coefficients in the outer solution. This gives a resonance condition to solve for the near extremal QNMs. This can be performed on either the RHS or the LHS of the contour, both giving the same QNMs. With this, we should be able to construct the radial solution everywhere in r. For clarity's sake, we will always refer to the two solutions as the inner and outer solutions, but then we take the near and far limits of these solutions.

1 Inner Solution

For the inner solution, we adopt the convention of [2] where we use the coordinates

$$r = r_+ + \epsilon\rho, \qquad t = 2M^2\tau/\epsilon, \qquad \phi = \psi + M\tau/\epsilon, \qquad a = \sqrt{1-\epsilon^2}. \tag{1}$$

Taking these coordinates, we can get the near-NHEK Teukolsky equation which is of the form

$$\rho(\rho+2)R''(\rho) + 2(\rho+1)(s+1)R'(\rho) + R(\rho)\left(\frac{(m\rho+m+\overline{\omega})(m\rho+m-2i(\rho+1)s+\overline{\omega})}{\rho(\rho+2)} + 2ims - \lambda_{lm}\right) = 0. \tag{2}$$

Here, $\lambda_{\ell m}$ is the normal Teukolsky angular separation constant and $\overline{\omega}$ is the correction to the QNM frequency given by

$$\omega = \frac{m}{2M} + \frac{\epsilon\overline{\omega}}{2M^2}. \tag{3}$$

A solution to this differential equation is

$$R_{\text{inner}}^{\text{in}}(\rho) = \frac{(i\rho)^{-\frac{1}{2}i(m+\overline{\omega})-s}(i(\rho+2))^{\frac{1}{2}i(\overline{\omega}-m)-s}}{2^{\frac{1}{2}i(\overline{\omega}-m)}e^{i\pi\left(-\frac{im}{2}-s\right)}}\,{}_2F_1\left[\alpha,\beta,\gamma;-\frac{\rho}{2}\right], \tag{4}$$

$$\alpha = 1/2 - s + i\delta - im \tag{5}$$

$$\beta = 1/2 - s - i\delta - im \tag{6}$$

$$\gamma = 1 - s - i(m+\overline{\omega}) \tag{7}$$

$$\delta^2 = m^2 - (s+1/2)^2 - \lambda_{\ell m}. \tag{8}$$

In general, the solution is a linear combination of the ingoing and outgoing solutions, but this Eq. (4) is just the ingoing solutions since everything needs to be ingoing at the horizon. Additionally, we take the either positive real or positive imaginary root for δ. This particular solution is valid for the inner region (including for complex r), is completely ingoing at the horizon, and is normalized to 1. The factors of i also ensure the branch cut at the outer and inner horizons ($\rho = 0, \rho = -2$) point upwards in the complex plane.

In general, near the horizon, the ingoing asymptotics are $R \sim \Delta^{-s}e^{-ikr_*}$. In these coordinates, the proper ingoing asymptotics are $\sim (\rho(\rho+2))^{-s}\rho^{-\frac{1}{2}i(m+\overline{\omega})}$ which is what we find (normalized to 1) when we take the limit of $\rho \to 0$. These asymptotics come from the fact that $r_* \approx \frac{1}{\epsilon}\ln(\rho\epsilon)$, $k \approx \frac{\epsilon}{2}(m+\overline{\omega})$ and the hypergeometric function ${}_2F_1 \to 1$ in this limit.

2 Outer Solution

In the outer region, we adopt Boyer-Lindquist-like coordinates. The only difference being instead of using r as our radial coordinate, we define

$$x = r - 1 - \epsilon. \tag{9}$$

We need to keep the $\mathcal{O}(\epsilon)$ term here to keep track of location of r_+. If we drop the $\mathcal{O}(\epsilon)$ term, the branch cut at the outer horizon won't coincide for the inner and outer solutions. Also, it's convenient to have the outer horizon to be at $x = 0$ which is precisely what we get in these coordinates. A more physical way to do this (and probably the way I would do it if I had to do it again) is to use something like $\tilde{x} = (r - r_+)/r_+$ which keeps proper track of the branch cut in a similar way. In these coordinates, the outer region Teukolsky Equation is

$$x\left(xR''(x) + 2(s+1)R'(x)\right) + \frac{1}{4}R(x)\left(-4\lambda_{\ell m} + m^2(x+2)^2 + 4imsx\right) = 0 \tag{10}$$

with a general solution of the form

$$R_{\text{outer}}(r) = Ae^{-\frac{1}{2}imx}(imx)^{i\delta-s-\frac{1}{2}}\,{}_1F_1\left[im - s + i\delta + \frac{1}{2}; 2i\delta + 1; imx\right] + B(\delta \to -\delta). \tag{11}$$

for $x \in \mathbb{C}$. The notation $(\delta \to -\delta)$ means taking the previous term and taking every δ to a $-\delta$. This implicitly means also taking every $\alpha \to \beta$. Next, we want to make sure this general solution is composed only of outgoing waves at spatial infinity ($x \to \infty$). The ingoing wave conditions should also be obeyed at $x \to -\infty$. To do this, we first need to make use of the asymptotics of the ${}_1F_1$'s.

3 Asymptotics

Here we discuss the asymptotic forms of both the inner and outer solutions under different limits.

3.1 Far Limit of Outer Solution

Coming from DLMF 13.7.1 [3], we have that as $z \to \infty$,

$$ {}_1F_1[a,b;z] \sim \Gamma[b]\left(\frac{e^z z^{a-b}}{\Gamma[a]} + \frac{e^{\pm i\pi a} z^{-a}}{\Gamma[b-a]}\right), \quad -\pi/2 < \pm \mathrm{ph}(z) < 3\pi/2. \tag{12} $$

With this phase convention for the $\pm$, the RHS of the branch cut in x is the $(+)$ asymptotic and the LHS is the $(-)$ asymptotic. In our particular case, we have that $z = imx$, and then you can think about the phases of z on both sides of the branch cut. Plugging this in, we get

$$ \begin{aligned} R_{\text{outer}}(x) = {} & A e^{-\frac{1}{2}imx}(imx)^{-\frac{1}{2}-s+i\delta}\Gamma[1+2i\delta] \\ & \times \left(\frac{e^{imx}(imx)^{-\frac{1}{2}-s-i\delta+im}}{\Gamma[\frac{1}{2}-s+i\delta+im]} + \frac{e^{\pm i\pi(im-s+i\delta+\frac{1}{2})}(imx)^{-\frac{1}{2}+s-i\delta-im}}{\Gamma[\frac{1}{2}+s+i\delta-im]}\right) + B(\delta\to-\delta). \end{aligned} \tag{13} $$

Now we must figure out what the desired asymptotics for the (lack of) ingoing waves at infinity. From Teukolsky I [4], $R^{\text{in}}(r\to\infty) \sim e^{-i\omega r_*}/r$. However, we must think about large r_* in terms of r. The general equation is

$$ r_* = r + \frac{2r_+}{r_+ - r_-}\ln(r-r_+) - \frac{2r_-}{r_+ - r_-}\ln(r-r_-). \tag{14} $$

In the $r \to \pm\infty$ limit, this becomes

$$ r_* = r + 2\ln(r). \tag{15} $$

Taking $\omega \to m/2$, the ingoing wave condition is

$$ e^{-i\omega r_*}/r \to e^{-\frac{im}{2}r} r^{-im}/r \approx e^{-\frac{im}{2}x}x^{-im}/x. \tag{16} $$

With this condition for ingoing waves, we want this piece to be 0 since QNMs produce only outgoing waves at infinity. Pulling out these pieces from Eq. (13), we get

$$ e^{-\frac{im}{2}x}(imx)^{-1-im}\left(A\frac{\Gamma[1+2i\delta]e^{\pm i\pi(1/2-s+i\delta+im)}}{\Gamma[1/2+s+i\delta-im]} + B(\delta\to-\delta)\right) = 0. \tag{17} $$

We set this to 0 because we don't want any ingoing waves at infinty, it should be completely outgoing. We can then solve for the ratio A/B. In doing so, we make use of the identity $\Gamma[1+z] = z\Gamma[z]$. Also, since we have different $\pm$ signs depending on which side of the branch cut, we take $\mathcal{R} = A/B$ using the $(+)$ asymptotic for the RHS of the branch cut, and $\tilde{\mathcal{R}} = \tilde{A}/\tilde{B}$ using the $(-)$ asymptotic for the LHS of the branch cut.

$$ \mathcal{R} = \frac{A}{B} = \frac{e^{2\pi\delta}\Gamma[-2i\delta]\Gamma[1/2+s+i\delta-im]}{\Gamma[2i\delta]\Gamma[1/2+s-i\delta-im]} \tag{18} $$

$$ \tilde{\mathcal{R}} = \frac{\tilde{A}}{\tilde{B}} = \frac{e^{-2\pi\delta}\Gamma[-2i\delta]\Gamma[1/2+s+i\delta-im]}{\Gamma[2i\delta]\Gamma[1/2+s-i\delta-im]}. \tag{19} $$

3.2 Near Limit of the Outer Solution

Now that we have the proper inner and outer functions, we need to perform the matching in the intermediate zone. Let's take the $R_{\text{outer}}(x\to 0)$ limit (near limit) first. When we do this, the exponentials and ${}_1F_1$'s tend to 1 in Eq. (11), leaving us with

$$ R_{\text{outer}}(x\to 0) = A(imx)^{-1/2-s+i\delta} + B(imx)^{-1/2-s-i\delta}. \tag{20} $$

I should have used different notation here, but this can be either for A, B or $\tilde{A}, \tilde{B}$, it doesn't change.

3.3 Far Limit of the Inner Solution

Now let's consider the asymptotics of Eq. (4) on the RHS of the branch cut. Using 15.8.2 of DLMF[1],

$$\frac{\sin(\pi(b-a))\,{}_2F_1[a,b,c;z]}{\pi\Gamma[c]} = \frac{(-z)^{-a}{}_2F_1\left[a,a-c+1,a-b+1;\frac{1}{z}\right]}{\Gamma[b]\Gamma[c-a]\Gamma[a-b+1]} - \frac{(-z)^{-b}{}_2F_1\left[b,b-c+1,b-a+1;\frac{1}{z}\right]}{\Gamma[a]\Gamma[c-b]\Gamma[b-a+1]}. \quad (21)$$

By taking the limit $-z\to\pm\infty = \rho\to\pm\infty$ so we're in the matching regions, we find the ${}_2F_1$'s go to 1 on the RHS of Eq. (21). Substituting this into Eq. (4) gives

$$R^{\text{in}}_{\text{inner}}(\rho\to\pm\infty) = \frac{(i\rho)^{-\frac{i}{2}(m+\overline{\omega})-s}(i(2+\rho))^{\frac{i}{2}(\overline{\omega}-m)-s}}{2^{\frac{i}{2}(\overline{\omega}-m)}e^{i\pi(-\frac{im}{2}-s)}}\cdot\frac{\pi\Gamma[\gamma]}{\sin(\pi(\beta-\alpha))} \times\left(\frac{(i\rho)^{-\alpha}e^{-\frac{i\pi\alpha}{2}\pm i\pi\alpha}2^{\alpha}}{\Gamma[\beta]\Gamma[\gamma-\alpha]\Gamma[\alpha-\beta+1]} - \frac{(i\rho)^{-\beta}e^{-\frac{i\pi\beta}{2}\pm i\pi\beta}2^{\beta}}{\Gamma[\alpha]\Gamma[\gamma-\beta]\Gamma[\beta-\alpha+1]}\right). \quad (22)$$

Note that some of the prefactor terms will cancel since $i\rho\sim i(2+\rho)$ in this limit. Also, since it is now convenient to work in x instead of ρ, we invert Eq. (1) and take $\rho=(r-1)/\epsilon-1=x/\epsilon$. We can also simplify the sin(...) term using the identity

$$\frac{\pi}{\sin(\pi z)} = \Gamma[z]\Gamma[1-z]. \quad (23)$$

Performing these transformations gives us the $x\to\pm\infty$ limit of the inner solution,

$$R^{\text{in}}_{\text{inner}}(x\to\pm\infty) = \frac{\Gamma[\gamma]}{2^{\frac{i}{2}(\overline{\omega}-m)}e^{i\pi(-\frac{im}{2}-s)}}\left(\frac{e^{-\frac{i\pi\beta}{2}\pm i\pi\beta}2^{\beta}(m\epsilon)^{1/2+s-i\delta}\Gamma[2i\delta](imx)^{-1/2-s+i\delta}}{\Gamma[\alpha]\Gamma[\alpha-\beta]} + \frac{e^{-\frac{i\pi\alpha}{2}\pm i\pi\alpha}2^{\alpha}(m\epsilon)^{1/2+s+i\delta}\Gamma[-2i\delta](imx)^{-1/2-s-i\delta}}{\Gamma[\beta]\Gamma[\beta-\alpha]}\right). \quad (24)$$

3.4 Matching Conditions and QNMs

From Eq. (24), we can read off the expression for A, B and $\tilde{A}$, $\tilde{B}$ by comparing to Eq. (20).

$$A = \frac{\Gamma[\gamma]\Gamma[2i\delta]e^{\frac{i\pi\beta}{2}}2^{\beta}(m\epsilon)^{1/2+s-i\delta}}{2^{\frac{i}{2}(\overline{\omega}-m)}e^{i\pi(-\frac{im}{2}-s)}\Gamma[\alpha]\Gamma[\gamma-\beta]} \quad (25)$$

$$B = \frac{\Gamma[\gamma]\Gamma[-2i\delta]e^{\frac{i\pi\alpha}{2}}2^{\alpha}(m\epsilon)^{1/2+s+i\delta}}{2^{\frac{i}{2}(\overline{\omega}-m)}e^{i\pi(-\frac{im}{2}-s)}\Gamma[\beta]\Gamma[\gamma-\alpha]}, \quad (26)$$

$$\tilde{A} = \frac{\Gamma[\gamma]\Gamma[2i\delta]e^{\frac{-3i\pi\beta}{2}}2^{\beta}(m\epsilon)^{1/2+s-i\delta}}{2^{\frac{i}{2}(\overline{\omega}-m)}e^{i\pi(-\frac{im}{2}-s)}\Gamma[\alpha]\Gamma[\gamma-\beta]} \quad (27)$$

$$\tilde{B} = \frac{\Gamma[\gamma]\Gamma[-2i\delta]e^{\frac{-3i\pi\alpha}{2}}2^{\alpha}(m\epsilon)^{1/2+s+i\delta}}{2^{\frac{i}{2}(\overline{\omega}-m)}e^{i\pi(-\frac{im}{2}-s)}\Gamma[\beta]\Gamma[\gamma-\alpha]}. \quad (28)$$

[1]We have to be quite careful with the phase conventions here. Eq. 15.8.2 of DLMF has phase conventions that makes you think it would only be valid for the RHS. However, if you use 15.8.5 which is definitely valid for the LHS and then use the equations in 15.4, you'll actually recover 15.8.2. This is very non-trivial and takes a lot of algebra. If you're not convinced, you can also just check it numerically for some values. The difference between the RHS and LHS is how we pull out the (-) from the $(-z)^{-a}$ and $(-z)^{-b}$ in terms of exponentials.

We can also ge the ratio of A/B and $\tilde{A}/\tilde{B}$ from this which we will call $\mathcal{R}_{\text{match}}$ and $\tilde{\mathcal{R}}_{\text{match}}$,

$$\mathcal{R}_{\text{match}} = \frac{\Gamma[2i\delta]e^{\pi\delta}(2m\epsilon)^{-2i\delta}\Gamma[\beta]\Gamma[\gamma-\alpha]}{\Gamma[\alpha]\Gamma[\gamma-\beta]\Gamma[-2i\delta]}\,, \tag{29}$$

$$\tilde{\mathcal{R}}_{\text{match}} = e^{-4\pi\delta}\tilde{\mathcal{R}}_{\text{match}}. \tag{30}$$

Comparing this to Eq. (18), we see that $\mathcal{R}$ and $\mathcal{R}_{\text{match}}$ (as well as the tilde counterparts) can only be equal if we are near the pole of one of the Γ's, namely

$$\Gamma[1/2+s+i\delta-im] = \Gamma[\gamma-\alpha] = \Gamma[1/2-i\delta-i\varpi] = \Gamma[-n-i\eta] \tag{31}$$

for some integer (overtone number) n and some small deviation η which encodes the distance from the pole. This leads to an expression for the QNM shift $\overline{\omega}$,

$$\overline{\omega} = -(\delta + i(n+1/2)-\eta). \tag{32}$$

Now we just need the specific form of η which we can get by comparing $\mathcal{R}$ and $\mathcal{R}_{\text{match}}$. Combining Eq. (18) and Eq. (29), we get

$$\mathcal{R}_{\text{match}} = \frac{e^{\pi\delta}(2m\epsilon)^{-2i\delta}\Gamma[2i\delta]\Gamma[\beta]\Gamma[-n-i\eta]}{\Gamma[\alpha]\Gamma[-2i\delta]\Gamma[-n+2i\delta-i\eta]} \approx \frac{e^{\pi\delta}(2m\epsilon)^{-2i\delta}\Gamma[2i\delta]\Gamma[\beta]i(-1)^n}{\Gamma[-2i\delta]\Gamma[\alpha]\Gamma[-n+2i\delta]n!\eta} \tag{33}$$

for small η. You can get this expansion for $\eta \ll 1$ in Mathematica using *Series* and the assumptions $n \in \mathbb{Z}, n \geq 0$. This gives us an expression for η in terms of $\mathcal{R}$ coming from the outgoing condition,

$$\eta \approx \frac{e^{\pi\delta}(2m\epsilon)^{-2i\delta}\Gamma[2i\delta]\Gamma[\beta]i(-1)^n}{\Gamma[-2i\delta]\Gamma[\alpha]\Gamma[-n+2i\delta]n!\mathcal{R}}. \tag{34}$$

If we plug in $\mathcal{R}$ from Eq. (18) into this, we get a more explicit expression for η,

$$\eta \approx \frac{(2m\epsilon)^{-2i\delta}e^{-\pi\delta}i(-1)^n\Gamma^2[2i\delta]\Gamma[1/2-s-i\delta-im]\Gamma[1/2+s-i\delta-im]}{n!\Gamma[-n+2i\delta]\Gamma^2[-2i\delta]\Gamma[1/2-s+i\delta-im]\Gamma[1/2+s+i\delta-im]}. \tag{35}$$

Here, I have done this for the RHS matching, but you can do the same matching on the LHS and you will get the same expression for $\overline{\omega}$ and η. Note that this differs slightly from Aaron's note which has $e^{+3\pi\delta}$ instead. I believe this is from a mistake in his equivalent to my Eq. (13) where he takes the LHS far asymptotics of the outer solution rather than the RHS asymptotics. This should be checked more thoroughly though.

With all this information, we can reconstruct the full solution everywhere. The inner solution takes the form of Eq. (4) with the transformation $\rho \to x/\epsilon$. Similarly, the outer solution takes the form of Eq. (11) with the $A, \tilde{A}$ and $B, \tilde{B}$ coefficients coming from Eqs. (25),(27) and (26),(28), with the shift in the QNM frequency, $\overline{\omega}$, coming from Eq. (32). If we then plot both these functions along the branch cut, they should match one another in the intermediate zone which I have found they do in *Mathematica*. One thing to note about the *Mathematica* implementation is that it will only work for $m > 0$. For $m < 0$, these modes have negative frequencies so they don't approach the horizon frequency so this approach won't work. For $m = 0$, a bunch of stuff becomes singular and a different approach is needed as well. Another thing to be careful for is that for $m/l \lesssim .74$ or whatever it is, if you make ϵ too small, η is actually incredibly tiny, like smaller than the machine precision of *Mathematica*. I have upped the working precision to what I think should be good for basically all cases, but you should be careful about the precision. This will occur for the 210 mode if $\epsilon \lesssim 10^{-5}$ for machine precision. However, for $\epsilon \gtrsim 10^{-5}$, the matching actually isn't great because I don't think you're close enough to extremality. If the precision isn't high enough the output of $\mathcal{R}$ and $\mathcal{R}_{\text{match}}$ they won't actually match because it's beyond the numerical precision. My implementation can be found on my GitHub here.

References

[1] Pablo A. Cano and Marina David. Teukolsky equation for near-extremal black holes beyond general relativity: Near-horizon analysis. *Physical Review D*, 110(6):064067, September 2024.

[2] Pablo A. Cano and Marina David. Teukolsky equation for near-extremal black holes beyond general relativity: Near-horizon analysis. *Physical Review D*, 110(6):064067, September 2024.

[3] *NIST Digital Library of Mathematica Functions.*

[4] Saul A. Teukolsky. Perturbations of a Rotating Black Hole. I. Fundamental Equations for Gravitational, Electromagnetic, and Neutrino-Field Perturbations. *The Astrophysical Journal*, 185:635–648, October 1973.